Housing First or Treatment First? Ask the Kids, Not the Parents

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Response to Referee 1

Comments in order, responses in red

Thank you for a detailed and careful report. You ordered your comments by importance as 2, 1, 4, 3. I answer them below in the order of the report, and I start with a short overview of what changed in the paper that bears on them.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First or Treatment First? Ask the Kids, Not the Parents*. The reason is the substance of the revision rather than a relabelling, and it should become clear over the course of this letter.

What changed that bears on your report. The version you read ranked rationed and universal programmes, and it put universal Treatment First ahead of universal Housing First only once a loss of consortium was added to welfare after the model was solved. Your comments 3(a) to 3(c) were right about that adjustment. It is gone, and so are the universal programmes. The new paper compares the two policies within the existing rationed programme, and it asks first whether moral hazard, the argument that dominates the policy debate, justifies a sobriety condition. It does not: with the externalities on children switched off, a newborn is worse off under Treatment First by 1.09 percent of lifetime consumption, even when the savings from the drug test are spent on more vouchers. The answer turns instead on the children of users, through three externalities that a parent counts only in part. A parent's drug spending, and the unemployment that a substance use disorder (SUD) brings, take consumption and housing from the child; a parent's SUD lowers the child's productivity as an adult; and a revoked voucher can put the child in a shelter. With these in the model, and with the drug price clearing its market and the chance of drawing a voucher adjusting to balance the budget, Treatment First cuts drug use by 21 percent and homelessness by 11 percent, and raises a newborn's welfare by 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count.

Many of your other comments are now answered in the model or in the calibration. The quantitative model reports its fit on every targeted moment (Tables 3 and A1) and its untargeted moments beside the data (Table 4). It now has serious mental illness (SMI), a treatment choice, a drug price that clears its market, a drug test that catches only some users and whose cost is paid from the budget, and a transition that is solved and reported (Sections 6.4 and 6.5). A two period stylized model with saving (Section 3) replaces the old static appendix, and the comparison you ask for in 1(d) is run in it. I also removed endogenous labour supply, rents, homeownership and house prices, which in earlier versions had a negligible effect on the results (Section 4.4, footnote 19); this bears on 2(c) and 2(d). Where the paper does not do what you asked, the response says so and is marked as still open.

1. Model

1(a) Mental health is largely absent

"Mental health is central to the economic and policy environment being studied, but it is largely

absent from the model. . . . It seems to me that mental health contributing to drug use and addiction as well as homelessness would weaken TF dramatically as a policy."

Added. SMI, mental illness severe enough to cause substantial functional impairment, is now a state of the model, a two state chain. Onset follows an age profile that reproduces a log linear fit to the share of adults with SMI at each age in the 2024 National Survey on Drug Use and Health (NSDUH), scaled by a level calibrated to its prevalence, 4.82 percent (4.83 in the model). A household with a SUD is twice as likely to develop SMI ($\iota = 2.00$), between the hazard ratios of 1.34 and 6.04 in the Danish registers (Hjorthøj et al., 2018; Nielsen et al., 2017). A household recovers from SMI with probability 0.066 a year, from the ten year recovery rate in Åsbø et al. (2022).

SMI acts through three channels, each set from its own source. It makes a job harder to find, with the divisor κ_m^{ue} calibrated to the SMI share of the homeless, 25 percent in Gutwinski et al. (2021) (24.7 in the model). It raises the baseline risk of dying 2.50 times (Olfson et al., 2015). And it lowers the chance of recovering from a SUD, by the multiplier 0.82 (Bradizza et al., 2006). So the link between mental illness and drug use runs both ways: a SUD raises the chance of SMI, and SMI makes a SUD more persistent. Untargeted, households with SMI are 2.64 times as likely as others to use drugs in the model, against 2.44 in NSDUH 2024.

Two choices are worth stating. SMI does not enter the utility function; it acts only through these three channels. And housing does not move the SMI chain, because in the At Home / Chez Soi trial housing had no detectable effect on the symptoms of homeless adults with SMI (Stergiopoulos et al., 2015).

On your expectation that mental illness weakens Treatment First: in the model it pulls both ways. Households with SMI use drugs and are homeless at higher rates than others, as in the data, so Treatment First's drug test falls disproportionately on them. For those it puts on the street, losing the voucher raises three year mortality by about 3.6 percentage points, of the same order as the 4.6 points by which Costantini (2025) finds Housing First lowers the three year mortality of homeless veterans with mental illness. But a SUD raises the onset of SMI, so deterring use also prevents some SMI, with the unemployment and the mortality that come with it. With SMI in the model and calibrated, Treatment First still raises a newborn's welfare by 2.57 percent. **[Still open.]** The paper does not report the model without SMI, so I cannot say how much SMI moves that number.

This is in Section 4.7, Block 3 of Section 5.1 (Table 3), and Blocks 2 and 7 of Section 5.2 (Table 4).

1(b) Treatment First contains no treatment

"One of the goals of TF as a policy is to encourage addicts to seek treatment for their addiction (it is even in the name of the policy!). However, this mechanism is ruled out in the model by the assumption that recovery is random. I found the argument for ruling out any possibility of this mechanism to be fairly weak given its centrality to one of the two policies studied."

[Conceded.] You are right, and treatment is now in the model. A household with a SUD chooses each year whether to be in treatment, and treatment doubles its annual chance of recovering ($g = 2.00$, from the remission rates under medication for opioid use disorder in Paul et al., 2024, and the trials of agonist medication in Fudala et al., 2003). Treatment costs the household utility, $\bar{\tau}^u$, calibrated to treatment take up: 24.5 percent of adults with a SUD received treatment in

the past year in NSDUH 2024, against 24.9 in the model. It costs the government $9,426 a year, the cost of a residential treatment episode in Roebuck et al. (2003), under both policies, since treatment is publicly funded. Recovery is also less likely while homeless and with SMI.

Treatment First now requires treatment, as the July 2025 executive order directs: a household with a SUD keeps its voucher only if it is in treatment and does not use. In the main experiment the share of households with a SUD who are in treatment rises from 24.6 to 29.0 percent.

On your expectation that this strengthens Treatment First: in the long run it is not what drives the result. When households with a SUD cannot keep their voucher through treatment at all, Treatment First cuts drug use and the share with a SUD by 29 percent each, homelessness falls by 11 percent, and a newborn gains at least as much as in the main experiment (footnote 38). Where treatment matters is the transition. Along the main path homelessness and the share of children in a shelter never rise, and one of the three reasons Section 6.4 gives is that few households lose their vouchers, since a household with a SUD keeps its voucher by entering treatment and not using. When households with a SUD cannot stop using, a test with no exemption from the first day raises homelessness by up to 14 percent if it catches every user, and by up to 5 percent if it catches one in five (Section 6.5).

This is in Sections 4.6 and 4.9, Block 4 of Section 5.1 (Table 3), Section 6.2 (Table 5) and Sections 6.4 and 6.5 (Table 7).

1(c) The drug price is fixed at one

> *"However, in the model, the drug price appears to be fixed at one and does not adjust in the policy counterfactuals. . . . this seems like a strong assumption that either should be relaxed or better defended."*

Relaxed. The drug price is now set in a market that clears against a supply curve with a constant price elasticity ε_s. The supply shifter is set so that the price is one in the benchmark and is then held fixed, so the price moves in any counterfactual that lets the market clear. It clears in the main experiment and in every period of the transition; other runs hold it at its benchmark level so that its effect can be read off on its own. The Associate Editor asks for the same thing, and Referee 3 asks how the price can be varied when it is normalised to one; this is the answer to both.

Your concern about offsetting is borne out, and the paper reports its size. In the main experiment the price falls by 5 percent, which draws some households back into use. With the price held and the chance of drawing a voucher, π_V, balancing the budget, Treatment First cuts drug use by 42 percent and a newborn gains 6.45 percent; with the price clearing as well, use falls by 21 percent and the gain is 2.57 percent.

The size of the offset depends on ε_s. I know of no direct estimate for an illicit drug. Cannabis is the closest market I could find: it remains a Schedule I substance federally, so its suppliers work under federal prohibition, as an illicit supplier does. I set ε_s to 1.46, the estimate of Mace et al. (2020) on Washington State administrative data. That is a short run elasticity, over a month; the long run elasticity of an illicit supply is likely higher, and studies of illicit drugs take it to be infinite (Becker et al., 2006). A more elastic supply means a smaller fall in the price and a smaller offset, so the offset in the paper is more likely too large than too small. The paper does not report the results at another value of ε_s. On the demand side, the model's participation elasticity, -1.32, is larger in absolute value than the estimates of about -0.3 to -0.9, which again works against Treatment First (2(f)).

This is in Section 4.12, Block 6 of Section 5.1 (footnote 31), Block 5 of Section 5.2, and Section 6.2 (Table 5).

1(d) Show that the mortality channel is the quantitatively important one

> *"... it seems possible that the labor-market and income-risk consequences of drug use could also generate a low-income concentration of drug use. The paper should demonstrate that the claimed mortality/value-of-life channel is actually the quantitatively important one. A simple way to do this would be to compare the relationship between income and drug use in the baseline model to an otherwise similar version of the model in which mortality is not affected by drug use."*

[New result.] Done in the stylized model of Section 3, which keeps the saving decision you were concerned about, and the saving decision turns out to be what decides between the two channels. **[Still open.]** It is not done in the quantitative model: the paper does not report that model solved again with drug use leaving mortality unchanged.

In the stylized model a household can save but not borrow, and a SUD raises both its risk of dying and its risk of unemployment. It is calibrated so that participation is 8.90 percent and the average user earns 0.74 of the population average, as in NSDUH 2024. Your comparison is then 0.74 with both channels at their calibrated strength, against 3.51 with mortality unaffected by a SUD and the unemployment channel at its calibrated strength. Section 3.3 and Figure 1 run it one channel at a time, from zero to twice its calibrated strength, with b recalibrated at each point so that participation stays at 8.90 percent and only who uses changes.

With neither penalty, use concentrates at the top, because the drug is a luxury good: the average user earns 3.51 times the population average. The mortality channel alone takes this to 0.64 at its calibrated strength, crossing the average at 73 percent of that strength, and to 0.27 at twice it. The unemployment channel alone leaves the average user at 3.51 times the population average, even at twice its calibrated strength: on its own it does not change who uses at all.

Your conjecture holds in one case, and the paper shows it. With saving switched off, the right panel of Figure 1, the unemployment channel also reverses the gradient, to 0.75 at its calibrated strength and 0.27 at twice it. The difference is self insurance. A household that can save insures itself against losing its job, so its loss is roughly proportional to its income and costs a rich household about as much in utility as a poor one. Nothing insures it against dying, which takes away the whole continuation value of being alive, and that loss grows with income. Without saving, losing a job takes a household down to the food assistance floor whatever it earned, so that loss grows with income too.

The result also holds in a version of the stylized model with housing, a shelter, unemployment insurance and housing vouchers. There, with no consequences of a SUD, the average user earns 3.54 times the population average; the risk of dying alone takes this to 0.52 at twice its calibrated strength; the risk of losing one's job alone leaves it at 3.54; and only with saving switched off does job loss on its own bring it down, to 0.56 (footnote 10).

In the quantitative model the SUD mortality multiplier, $\kappa_{\phi d}$, is the parameter calibrated to the income of users, for the reason Section 3 gives, and Figure 2 compares the model's gradient with NSDUH 2024 (2(c)(i)).

This is in Sections 3.1 to 3.3 (Tables 1 and 2, Figure 1, footnote 10), Block 1 of Section 5.1 (Table 3) and Section 5.2 (Figure 2).

1(e) The appendix is disorganised and claims more than it establishes

"... the appendix is useful for developing intuition, but I found it somewhat disorganized and narrower than the surrounding discussion suggests. The results are derived in very specific static settings, while the main model is dynamic ... the appendix should be reorganized and made more precise and the main text discussion should describe it more clearly as intuition rather than as a general proof of the model's mechanism."

[Conceded.] Accepted. The static appendix is gone. In its place, Section 3 is a stylized model in the body of the paper: two periods of thirty years, a saving decision, a decision to use, a SUD that raises the risk of dying before the second period and of unemployment in it, and food assistance as the only transfer. It is calibrated (Tables 1 and 2) and hits its five targets exactly. The paper says that its sole purpose is to show why the income gradient in drug use reverses (1(d)). The welfare results come from the quantitative model of Section 4, which the paper introduces as the model its numbers come from. The Associate Editor asked for the same section.

Three footnotes say how far the result travels: it holds with log utility (footnote 8), when the risk of a SUD rises with use (footnote 9), and in a version with housing, a shelter, unemployment insurance and vouchers (footnote 10). The only appendix now is Appendix A, which sets out the standard parameters. The paper does carry the mechanism over to the quantitative model, where it is the reason the SUD mortality multiplier is calibrated to users' income, but it does not test the mechanism there (1(d)).

This is in Section 3.

2. Calibration and quantification

2(a) Model fit never reported

"... a notable omission is the model fit. That is, the author never reports the model's moments even though they would naturally fit in the table. The closest that the author comes to reporting these moments are by saying, 'My benchmark closely matches all calibration targets.' This is incredibly vague."

[Conceded.] You are right, and I agree with 2(b) that without the model's moments the fit could not be checked. Table 3 now has a Data and a Model column for each of the 26 parameters that carry the paper's results, and Table A1 does the same for the 33 standard ones. For every parameter calibrated internally, 21 in all, the Model column gives the benchmark's value of its target beside the data. The largest gaps are the income of users, 62 percent of the average in the model against 74 percent in the data (2(c)(i)); renters paying more than half their income, 25.8 against 19.2 percent; and the unemployment rate, 5.6 against 4.9 percent. The stylized model of Section 3 hits all five of its targets exactly (Table 2). The sentence you quote is gone.

This is in Section 5.1 (Table 3) and Appendix A (Table A1).

2(b) Untargeted misses framed too positively

"I find these tables to broadly demonstrating that the model misses substantially across the most important dimensions presented. ... most problematic in my eyes is that the author frames the model as having done a reasonably good job of matching these moments. ... Without being able to verify myself ... I cannot help but question both the fit of the calibration and also the entire quantification exercise."

[Conceded.] Accepted, and the sentence you quote is gone. Section 5.2 now reports every untargeted comparison in eight blocks (Table 4), with the model's number beside the data's. The main misses are these. The homeless in the model are too concentrated at ages 35 to 44, too few of them are 55 and over, and fewer of them work or have income (Block 1). Users are younger than in the data, and excess deaths due to a SUD are a smaller share of deaths at ages 25 to 44 than drug poisonings are in the data (Block 2). More housed households spend over thirty percent of their income on rent than renters do in the data, and voucher holders pay a larger share of their income in rent (Block 3). The bottom half of households holds a larger share of net worth than in the data (Block 4). Participation responds more to the drug price than in the estimates (Block 5). And homelessness passes from parent to child more strongly than in Bassuk et al. (1997) (Block 6).

Where the paper can sign what a miss does to the result, it says so. A larger price response can only work against Treatment First, since the lower drug price under Treatment First draws more households back into use (Block 5). With transmission at the low end of the evidence on the SUD and above it on homelessness, the model loads the externality on children towards Housing First more than Treatment First (Block 6). For the other misses the paper does not sign the effect. With the fit of every target now in Tables 3 and A1 (2(a)), you can check both the targeted and the untargeted fit yourself.

This is in Section 5.2 (Table 4).

2(c)(i) Homeless and users are too poor and too detached from work

> *"In the model, all homeless individuals are unemployed, compared with 40 to 52 percent in the data, and average income among the homeless is essentially zero, compared with 19.1 percent of average income in the data. Similarly, the average income of drug users is 41 percent of average income in the model, compared with 74 percent in the data. . . . a model in which these groups are much poorer and less attached to employment than in the data may evaluate HF and TF at the wrong margins."*

[Conceded.] You are right, and the paper now targets the 74 percent you cite. In both models a user is a household whose use of an illicit drug amounts to a disorder, mild included, the counterpart of adults aged 26 and over with a drug use disorder of any severity in NSDUH 2024, whose average income is 74 percent of the population average. A household with a SUD is one whose disorder is moderate or severe, and use that never reaches a disorder is not modelled (footnote 26). The stylized model of Section 3 matches the 74 percent exactly. In the quantitative model the SUD mortality multiplier, $\kappa_{\phi d}$, is calibrated to it, and the model reaches 62 percent (Table 3). Users in the model are still poorer than in the data, but by much less than in the version you read.

The calibration targets only the average income of users. Figure 2, new in Section 5.2, compares the whole distribution of use by income with NSDUH 2024 and shows where the remaining gap comes from. As in the data, use is highest at the bottom and lowest at the top: 18.3 percent of households with less than 0.22 of the mean income use drugs, against 13.7 percent in the data, and 4.2 percent of those with more than 1.62 times the mean, against 4.8 percent. Between 0.43 and 1.62 times the mean, use in the model is flatter and lower than in the data, 5.9 to 6.3 percent against 7.2 to 10.2. Both differences lower the average income of users in the model.

The homeless are also less deprived than in the version you read, though still more than in the data. Of the homeless in the model, 21.4 percent are employed, against 46 percent in Meyer et al. (2024), and their income is 6.5 percent of the average, against 19.1 percent. Both are untargeted.

On the margins you list, Section 5.2 now says why the income of users matters for the experiments: the vouchers are means tested, so the lower in the income distribution users sit, the more of them are eligible for a voucher, and the more of them, and of their children, Treatment First's test reaches. The model no longer has a labour supply choice, so employment works through the chances of losing and finding a job, which depend on productivity, homelessness, a SUD and SMI; untargeted, unemployment among households with a SUD is 14.5 percent in the model against 16.2 percent in NSDUH 2024. And the rise in tax revenue under Treatment First comes mainly from children who start adult life with higher productivity, and only to a lesser extent from fewer adults out of work because of a SUD.

[Still open.] The misses remain, and the paper does not show how the results would change if users and the homeless were as well off as in the data.

This is in Block 1 of Section 5.1 (Table 3), Blocks 1 and 2 of Section 5.2 (Table 4, Figure 2), Sections 4.3 and 4.8, and Section 6.2.

2(c)(ii) The model's homeless are too young

> *"In the model, 73.1 percent of homeless individuals are between ages 26 and 44, compared with 48.1 percent in the California data. Conversely, only 9.3 percent of the model's homeless population is age 55 or older, compared with 27.1 percent in the data. . . . If the model shifts homelessness toward younger ages, it may overstate the long-run labor-market and addiction-response channels while understating the role of policies for older homeless individuals."*

[Conceded.] You are right, and the miss has narrowed but not closed. The age distribution of the homeless is now compared with the national point in time count of the Department of Housing and Urban Development for 2024. The youngest band, 25 to 34, is the target of the endowment at entry to adulthood, T_0 (26.4 percent in the model against 26.0 in the data); the other four bands are untargeted. The model still has too many of the homeless at 35 to 44, 37.9 percent against 27.2, and too few at 55 and over, 11.5 percent against 25.9. Users, too, are younger in the model than in the data (S1).

Section 5.2 now says why the age of users matters for the experiments: both policies reach a child through its parent's decisions, so the age of users decides how many children their use touches. The paper's check on how many children the externalities reach is the share of children living with a parent who has a SUD, 5.8 percent in the model against 4.6 percent in NSDUH 2024 (Block 6). **[Still open.]** The paper does not show how the results would change with an older homeless population, and I do not claim the gap is harmless.

This is in Block 2 of Section 5.1 (Table 3) and Blocks 1, 2 and 6 of Section 5.2 (Table 4).

2(d) Match the policy relevant elasticities

> *"Relevant objects include the response of drug use to voucher generosity or conditionality, the response of homelessness to voucher access, the response of labor supply and employment to vouchers, and the response of recovery or addiction persistence to housing status. . . . the RCT literature cited in the introduction may provide identified elasticities that the model can match."*

Of the responses you list, two are now matched or compared with an estimate: the response of recovery to housing status, which is a calibration target, and the response of drug use to conditionality, through two estimates. The paper adds a third, the response of mortality to losing a voucher. The other three are not compared with an estimate.

- *Recovery and housing status.* μ_H, the multiplier on the chance of recovering from a SUD while homeless, is calibrated to Milby et al. (2005), who find that housing improves the chance of recovering by 47.3 percent (1.47 in the data, 1.45 in the model).
- *Drug use and conditionality.* Milby et al. (2005) also randomise housing conditional on abstinence against housing without a condition, among homeless adults with a cocaine disorder in treatment: 8.32 against 6.60 consecutive weeks of abstinence, a ratio of 1.26. The model's counterpart, the recovery hazard of voucher holders with a SUD under the condition against that of housed households with a SUD and no voucher, in the first year, is 1.40, untargeted.
- *Deterrence of a sobriety condition.* The detection rate of Treatment First's drug test is set so that the test deters use as much as in the closest evidence, the US military's random testing with dismissal, after which service members who stayed were 61 percent less likely than civilians to have used marijuana in the past year; counting those who left, the fall was 11 to 45 percent (Mehay and Pacula, 1999). In the model, 62 percent of the users who face the test stop when it is introduced (4(a)).
- *Housing and mortality.* For households with SMI whom Treatment First's test puts on the street, losing the voucher raises three year mortality by about 3.6 percentage points in the model, with the drug price and π_V held, against the 4.6 points by which enrolling in Housing First lowers the three year mortality of homeless veterans with mental illness in Costantini (2025), untargeted. The paper compares the model with Costantini's Housing First estimate only; the finding that the Treatment First programmes of the Department of Veterans Affairs have no lasting effect on mortality is cited in Section 2 but not compared with the model.
- *Labour supply and employment.* The model has no labour supply choice: earlier versions had one, and it had a negligible effect on the results (footnote 19). Housing affects employment through κ_H, set at 0.20 from Desmond and Gershenson (2016): in a shelter, separations are $1 + \kappa_H$ times as frequent and job finding $1 - \kappa_H$ times as fast.

[Still open.] The paper does not compare the model's response of drug use to voucher generosity, of homelessness to voucher access, or of employment to vouchers with an estimate, although the Family Options Study it cites in Section 2 measures the effect of an unconditional subsidy on later homelessness (Gubits et al., 2016, 2018).

This is in Blocks 2 and 4 of Section 5.1 (Table 3), Blocks 7 and 8 of Section 5.2 (Table 4), the paragraph on the screen at the start of Section 6, and Section 4.4.

2(e) The price elasticity moment is not replicable

> *"... I found the description of the calculation of this moment to be extremely vague, to the extent that I would not be able to replicate it. This should be clarified. I would appreciate an appendix describing the model definitions of the moments more thoroughly in general."*

The participation elasticity is no longer a calibration target, so the calibration does not depend on how it is computed. The model's participation and intensive elasticities are reported as untargeted moments, -1.32 and -0.63, against about -0.3 to -0.9 and -0.3 to -0.8 in the literature. The response of drug use to Treatment First's test itself, which the paper keeps separate from the response to the price, is matched instead to the military's random testing (2(d), 4(a)).

On the general request: Tables 3, 4 and A1 give, for each parameter and each check, the moment and its source, and the notes to Table 4 and to Figure 2 define the model's counterpart where it is not a simple share. The replication package carries every dataset and a data appendix,

`data/DATA_APPENDIX.md`, which documents the dataset, year, universe and computation behind every data estimate. **[Still open.]** The paper has no appendix that defines the model's moments, and it does not spell out how the model's two price elasticities are computed.

This is in Block 5 of Section 5.2 (Table 4) and the opening of Section 5.

2(f) Show identification numerically, and show robustness to the elasticity range

> *"I did find the identification explanation in Section 4.2 to be quite helpful. . . . I would like this to also be shown numerically. This could be done in many ways: reporting the Jacobian of the moment vector with respect to the parameter vector, reporting the sensitivity statistic from Andrews, Gentzkow, and Shapiro (2017), through a figure showing how perturbing parameters locally around the baseline calibration affects the associated moments. . . . take the elasticity of drug use with respect to price which has a range of plausible estimates from −0.36 to −0.9 . . . whether or not the results are robust to this choice . . . should be demonstrated."*

[Still open.] The paper does not report a Jacobian, the statistic of Andrews, Gentzkow and Shapiro (2017), or a perturbation figure. What it does is narrower. Each parameter's paragraph in Section 5.1 says what the parameter controls, what happens as it rises and which moment it is matched to, in that order, and Table 3 now reports the model's value of each moment beside the data (2(a)).

On the price elasticity: you are right that its range is wide and that it matters for the policy result. It is no longer a target (2(e)), so the calibration does not depend on where in the range the target is placed. The model's participation elasticity, −1.32, is above the range in absolute value, and the paper argues that this is conservative: a larger price response can only work against Treatment First, since the lower drug price under Treatment First draws more households back into use. The paper does not solve the model with an elasticity inside the range.

For the other inputs that are measured imprecisely, the robustness checks the paper reports are listed under 4(d).

This is in Sections 5.1 and 5.2 (Tables 3 and 4).

3. Policy analysis

3(a) Loss of consortium is not integrated into the model

> *"If premature deaths impose welfare losses on other individuals, then these losses should in principle affect the value of life and agents' decisions (such as the decision to use drugs). Instead, the paper adds the loss-of-consortium effect only after solving the model. I find this to be an internal inconsistency."*

[Conceded.] You are right, and the adjustment is gone: nothing is added to welfare after the model is solved. I did not build a loss of consortium into the model; I dropped it. Referee 2 pointed to two effects on others that the previous version missed, a disamenity from homelessness and parents' concern for their children. The paper builds a version of the second, inside the household's problem.

Each household has a child, who lives with it for its first 25 years, from the parent's age 26 to 50, shares its consumption and dwelling but not the drug, and starts adult life with lower productivity

if the parent had a SUD while the child lived at home. The parent values its child at the weight ω_c, both in the child's flow utility during childhood (equation 12) and through what the parent's SUD costs the child's value as an adult, charged when the child leaves home (equations 24 and 25). So the child's welfare enters the parent's choices, including the choice to use, as you argue a loss to others should; only the share $1 - \omega_c$ that the parent leaves out is an externality. The welfare criterion counts the child's childhood at its own weight, and Section 6.3 measures the part of the gain the parent does not count (Table 6).

The disamenity is not in the benchmark. I solved the model with one, at the value in Loubière et al. (2020), about 0.2 percent of average earnings a year, and no long run gain in Table 5 moves by more than 0.14 percentage points.

This is in Sections 4.2, 4.8 and 4.10, Sections 6.1 and 6.3 (Table 6), and footnote 4.

3(b) The adjustment is ad hoc and a truly conservative value is zero

> *"κ_{cons} was plucked out of thin air, yet makes an enormous impact on the final policy rankings. To play devil's advocate on the point that the chosen value was conservative, a truly conservative value for a parameter plucked out of thin air such as this would be zero, which results in a different policy recommendation."*

[Conceded.] You are right, and the parameter is gone. The paper's first experiment is the case you describe, with no cost of a household's drug use to anyone else beyond the budget. The children are removed from welfare and from the parent's preferences, and the child's productivity no longer depends on what its parent did, so a household's drug use can harm only itself and the budget, and only moral hazard is left to justify a condition. In that setting Treatment First loses: a newborn is worse off by 1.17 percent of lifetime consumption with the drug price and π_V held, by 1.09 percent when π_V rebalances the budget, and by 0.74 percent when the drug price clears as well. The paper states this as a result: moral hazard alone does not justify conditionality.

The case for Treatment First then rests on the externalities on children, and the two parameters that set them are each tied to evidence. The child's earnings penalty, $\delta_d = 0.24$, is the estimate of Nilsson (2017), to my knowledge the only natural experiment on how a parent's substance shock affects a child's adult earnings. It measures exposure to alcohol in the womb, not a parent's disorder after birth, which no study measures directly. As an indirect check, the model's odds that a child of a parent with a SUD acquires one, 1.46, are at the low end of the 1.44 to 2.51 in the literature, which suggests that the penalty, if anything, is at the low end as well. The parent's weight on its child, $\omega_c = 0.085$, is set by behaviour: in NSDUH 2024, unmarried adults living with one other adult and no child are 17.5 percent more likely to have used illicit drugs in the past year than lone parents with one child, reweighted to the same age, sex, marital status and family income, and ω_c is the weight at which removing a parent's concern for its child raises use in the model by as much (17.7 percent).

The paper reports that the sign of the result survives a change in either. With ω_c at 0.69, the upper end of the literature (Nishiyama, 2000), a newborn gains 3.18 percent, though only 23 percent of that gain is then an externality; with the earnings penalty set to zero, Treatment First still raises welfare; and the earnings externality is 18 percent of the gain in the main experiment.

This is in Section 2, Section 6.2 (Table 5), Block 5 of Section 5.1 (Table 3), Block 6 of Section 5.2 (Table 4), and footnotes 5, 28 and 30.

3(c) Tone down the language

"Universal TF is not unambiguously preferred under all welfare criteria considered in the paper. . . . An example of language that should be softened is the conclusion's statement that the debate is resolved in favor of universal TF."

[Conceded.] Accepted. The paper no longer compares universal programmes, and it does not claim the debate is resolved. Every experiment keeps the programme rationed, as it is, and the answer is stated conditionally: Section 7 says that housing assistance should require sobriety if the externalities on children are counted, and that moral hazard alone is not a reason for it.

The paper also says who loses. As the trials imply, most households with a SUD are worse off under Treatment First: 74 percent of those who have one when the reform lands, with the drug price and π_V held as in a trial, and 67 percent when both clear. And the path to the long run depends on how the reform is designed (3(d)).

This is in Section 7, Section 6.4 and Section 6.5 (Table 7).

3(d) Solve proper transition paths

"The author should solve for proper transition paths. . . . I find the current strategy, of a several-period PE adjustment, to be a half-measure that is limited in its informativeness and opens up more questions than it answers."

[New result.] Done. The quantitative model's transition from Housing First to Treatment First is solved and reported. The reform lands as an unexpected permanent change at the start of year 1, the path starts from the benchmark, and the drug price and π_V clear the drug market and the budget in every period, for sixty years, to the Treatment First steady state.

You expected the slow adjustment of the SUD to matter, and it does, though not in a way that reverses the result. Drug use falls on impact, 81 percent of its long run fall in the first year. The share with a SUD falls only as fast as recovery allows, half of its long run fall by year 11 and nine tenths by year 27, and the drug price stays above its long run level for 33 years. Homelessness and the share of children in a shelter never rise: homelessness falls from 0.62 to 0.60 percent of households by year 10, and the share of children in a shelter from 1.03 to 1.00 percent. Section 6.4 gives three reasons. Few households lose their vouchers: at most 0.13 percent of households are excluded from the programme in any year, because a household with a SUD keeps its voucher by entering treatment and not using, and the test catches a user with a probability of only 20 percent a year. Drug use falls from the first year. And after the first three years π_V stays above its Housing First level. Children born at the reform gain 1.45 percent of lifetime consumption, less than the 2.57 percent of those born in the long run, because they grow up while more parents still have a SUD and before π_V has fully risen. Of the adults alive at the reform, 11 percent are worse off.

The design of the reform matters for the first years, and Section 6.5 shows how. When households with a SUD cannot stop using, a test with no exemption from the first day that catches every user raises homelessness by up to 14 percent and the share of children in a shelter by up to 11 percent, and they are above their Housing First levels in periods 1 to 15 and 2 to 11. If the test catches one in five users, the rises are up to 5 and 3 percent. Announcing the test five periods ahead lowers the rise but does not remove it. Exempting households in treatment for the first five periods keeps the share of children in a shelter from ever rising above its Housing First level, and holds the rise in homelessness to 1.6 percent, when the test catches one user in five. Children born at the reform gain in every one of these alternatives.

This is in Section 6.4 (Figure 3) and Section 6.5 (Table 7).

4. Sensitivity

4(a) Imperfect detection

> *"TF assumes that drug use is perfectly observed and perfectly enforceable. . . . If drug use were imperfectly observed, this would likely reduce the effectiveness of TF. The paper should analyze the sensitivity of the results to imperfect detection."*

Added. Under Treatment First every income eligible voucher holder, and every income eligible household that draws a voucher, is tested, and one that fails the condition is caught with probability p_{det}; with probability $1 - p_{\text{det}}$ it keeps its voucher. A household that is caught loses the voucher from the next period and is excluded from the programme for three years in expectation, the period for which the Housing Choice Voucher programme already bars a household evicted for drug related criminal activity.

p_{det} sets how strongly the test deters, so I set it to the closest evidence, the military's random testing (2(d)). At $p_{\text{det}} = 20$ percent, 62 percent of the users who face the test stop when it is introduced, against the 61 percent of Mehay and Pacula (1999) among service members who stayed. Enforcement is far from perfect: the test catches only 20 percent of users a year.

The sign of the result survives weaker detection. At half the rate, 10 percent, 41 percent of the users who face the test stop, inside the range of 11 to 45 percent that Mehay and Pacula find once those who left the military are counted. In the long run Treatment First then cuts drug use by 14 percent and homelessness by 6 percent, against 21 and 11 percent at the calibrated rate, and a newborn gains 1.65 percent, against 2.57. At the other end, Section 6.5 compares catching every user with catching 20 percent, along transitions in which households with a SUD cannot stop using (Table 7).

This is in Section 4.9 and the paragraph on the screen at the start of Section 6 (footnotes 33 and 35).

4(b) False positives

> *". . . the paper should consider the consequences of false positives. If some non-users lose voucher access because of erroneous tests, then TF may impose substantial costs on individuals who are not violating the sobriety requirement."*

[Still open.] Not added. In the model a household that does not use, and is in treatment if it has a SUD, is never flagged, because a positive test is confirmed in a laboratory before it has consequences. That is the paper's reason for leaving false positives out. It does not cover the other ways a household that complies could lose its voucher, such as administrative error, and the paper does not price those.

This is in the paragraph on the screen at the start of Section 6.

4(c) Compliance costs, stigma, administrative burden

> *"I found the argument that drug tests would cost only billions of dollars per year to implement to both be most likely a massive underestimation that ignored administrative overhead and assumed far*

more efficiency than is warranted of government policy, and also to be too large a number to entirely ignore in the first place."

You are right that the cost should not be ignored, and it is no longer outside the model. Each test costs the government $35, 0.031 percent of average earnings, the figure in the Texas Legislative Budget Board's fiscal note for Senate Bill 11, which is the upper end of the estimates, and every income eligible holder and applicant is tested (4(a)). When π_V adjusts to hold the government's net revenue at its benchmark level, as in the main experiment, every dollar spent on testing is a dollar less for vouchers.

Treatment, which Treatment First requires of households with a SUD, is a second cost. The government pays $9,426 a year for each household in treatment, under both policies, and as take up rises under Treatment First, part of what the test saves goes on treatment rather than on vouchers. The household bears a cost too: treatment costs it utility, and that cost is calibrated to observed take up.

[Still open.] What the paper does not do: it has no stigma, no administrative cost beyond the cost per test, and no sensitivity run on the cost per test.

This is in Sections 4.9 and 4.11, Block 4 of Section 5.1, and the paragraph on the screen at the start of Section 6.

4(d) Sensitivity to hard to identify parameters

"The paper should report sensitivity to parameters that are critical for the policy analysis yet are difficult to identify such as the parameters governing addiction transitions, recovery probabilities, addiction-related mortality, homelessness-related mortality, and the employment consequences of addiction and homelessness."

[Still open.] The paper does not report the results with any of these parameters changed. What it reports for each is where the value comes from and how the model compares with independent evidence.

- *Addiction transitions.* π_d is calibrated to the share of users with a SUD, 41.3 percent in NSDUH 2024 (42.6 in the model). In the stylized model the results hold when the risk rises with use (footnote 9), and in earlier versions of the quantitative model letting it rise with use had little impact on the results (footnote 20).
- *Recovery.* The baseline rate, 0.025 a year, is the hazard implied by the ten year remission rate in Vaillant (1995); the effect of housing, μ_H, is calibrated to Milby et al. (2005); and the effect of treatment, $g = 2.00$, is from Paul et al. (2024) and Fudala et al. (2003). The model's response of recovery to a sobriety condition, 1.40, is close to Milby's 1.26, untargeted (2(d)). And the long run result does not depend on treatment: when households with a SUD cannot respond at all, a newborn gains at least as much (footnote 38).
- *SUD mortality.* $\kappa_{\phi d} = 4.00$, calibrated to the income of users, lies between the mortality rate ratios of 3.91 in Plana-Ripoll et al. (2019) and 6.6 in Heiberg et al. (2018). Untargeted, excess deaths due to a SUD are 14.5 percent of deaths at ages 25 to 44 in the model, against 20.0 percent from drug poisonings in the Centers for Disease Control and Prevention (CDC) WONDER database.
- *Homelessness mortality.* $\kappa_{\phi H}$ multiplies baseline mortality by 1.60 at every age, the ratio for homeless adults against poor housed adults in Meyer et al. (2023). Untargeted, for households

with SMI whom the test puts on the street, losing the voucher raises three year mortality by about 3.6 percentage points, against Costantini's 4.6 (2(d)).

- *Employment.* κ_d^{ue} is calibrated to the SUD share of the homeless, 21.7 percent in Gutwinski et al. (2021) (20.2 in the model), and unemployment given a SUD is 14.5 percent in the model against 16.2 in NSDUH 2024, untargeted. The employment penalty of homelessness, $\kappa_H = 0.20$, is from Desmond and Gershenson (2016).

The robustness checks the paper does report are on the detection rate (halved, footnote 35), the parent's weight on its child ($\omega_c = 0.69$, footnote 28), the child's earnings penalty (set to zero, footnote 5), a disamenity from homelessness (footnote 4), and whether households with a SUD can respond to the condition at all (footnote 38 and Section 6.5). In each, Treatment First still raises a newborn's welfare.

This is in Section 5.1 (Table 3), Section 5.2 (Table 4) and the footnotes named above.

Smaller comments and typos

S1 Why a maximum experimentation age?

> *"I do not understand why there is assumed to be an age at which drug experimentation will not occur. This seems inconsistent with Table 4(a), which shows U-shaped drug use by age."*

[Still open.] The limit stays in the model, and the paper explains why. It is at model age 55, age 80 in a life that runs to 85, and it stops only households without a SUD from starting to use; a household that has a SUD at that age can keep using. So it limits when use can begin, not how long it lasts, and it binds only after age 80. Without it, using at old ages has little or no consequence, because too little life remains for the disorder to damage, and the model produces too much use in old age. The cause is that death is certain at the last age in the model, so close to it a household knows how little life remains, whereas in real life most people do not know when they will die.

The age profile of users is now reported against NSDUH 2023, and the model misses it at both ends. Users in the model are younger than in the data: 67 percent of them are 26 to 34, against 45 percent, and their mean age is 34.6 years, against 39.6. At 55 to 64 and at 65 and over the model has 5 and 2 percent of users, against 12 and 10 percent in the data.

This is in Section 4.6 (footnote 21), Block 13 of Appendix A (Table A1) and Block 2 of Section 5.2 (Table 4).

S2 Tables should show levels and percentage changes

> *"The results tables would be easier to interpret if they included both percentage changes and baseline/counterfactual levels for the main policy outcomes."*

Done in part. Table 5 reports, for every outcome, its level under Housing First and under each version of Treatment First side by side, with and without the externalities, and Section 6.2 gives the percentage changes in the text. Table 7 reports the transitions as changes against Housing First: the largest percent rise in homelessness and in the share of children in a shelter, and the

periods in which each is above its Housing First level. **[Still open.]** Table 5 does not have a separate column of percentage changes.

This is in Section 6.2 (Table 5) and Section 6.5 (Table 7).

S3 to S10 Typographical corrections

> *"well-aware" should be "well aware"; equation (9) should end with a comma; remove the comma after the citation in calibration target 19; footnote 7 "our calibration" should be singular; the Abramson citation lists forthcoming twice; "Kinneruda" and "Kragh-Sørensena" carry spurious trailing letters; Goering et al. has "Guido. Powell ,"; Degenhardt et al. has "Hamamd Ali" for "Hammad".*

The paper has been rewritten, and none of these passages remains: the sentence with "well aware", the old equation (9), the old calibration target 19 and the old footnote 7 are gone, and the paper is in the first person singular throughout. In the references, Abramson (2026) is listed once, as published online in the *Journal of Finance*, Goering et al. (2014) no longer carries the error, and Karlman et al. and Degenhardt et al. are no longer cited.

This is in the references.